

VIDSUM-GNN: Multimodal Video Summarization via Graph Neural Networks

1. Project Overview

VIDSUM-GNN is an AI-driven system for automatic video summarization. Its core goal is to convert long, unstructured videos into a short, meaningful representation that preserves key events and context. Unlike simple summarizers that rely only on text or only on visuals, this project is designed around multimodal understanding—it jointly considers visual content, audio/speech, and temporal structure to decide what is important and how to present it.

2. Theoretical Challenges: Why Video Summarization is Hard

Videos contain far more information than text, leading to several technical hurdles:

- **High Redundancy:** Many frames are nearly identical. Long videos include repeated scenes, pauses, and transitions.
- **Multiple Information Channels:** Meaning is derived from visuals (slides, objects, actions), audio (speech, emphasis), and timing.
- **Temporal Dependency:** A frame or scene may only make sense in the context of what happened before or after.
- **Ambiguity of "Importance":** What matters depends on user needs (e.g., a sports highlight vs. a lecture summary).

Summarization requires not just compression, but reasoned selection and structuring.

3. The Core Concept: Video as a Graph

A key theoretical design choice in VIDSUM-GNN is to treat a video not just as a sequence, but as a **graph of segments**.

1. **Segmentation:** The video is broken into **shots** (continuous segments between scene cuts). Each shot becomes a **node**.
2. **Edge Construction:** Connections between shots model relationships:
 - **Temporal Edges:** Link nearby shots in time to preserve flow and continuity.
 - **Semantic Edges:** Connect shots similar in meaning (topic continuity, repeated concepts).
 - **Audio/Speech-Informed Relations:** Capture cases where spoken content relates distant visual segments.

4. Technical Foundation: Graph Neural Networks (GNNs)

While CNNs or Transformers handle local or sequential data, VIDSUM-GNN uses an attention-based graph model (Graph Attention Network / GAT style) to handle non-local relationships.

Importance Scoring

The GNN aggregates information from a node's neighbors and propagates context across multiple "hops." The model learns attention weights—how much influence each neighbor should have—when computing a node's **importance score**.

Multimodal Features

To decide importance, each shot is represented by:

- **Visual Features (Vision Transformers, ViT):** Encodes appearance, layout, and key objects.
- **Audio & Speech (ASR via Whisper):** Converts speech to text for semantic reasoning.
- **Auxiliary Signals:** Motion intensity, audio energy, and scene change strength.

5. Methodology: The 9-Stage Pipeline

The system operates through a structured multistage pipeline:

1. **Input & Preprocessing:** Format normalization and audio extraction.
2. **Shot Detection:** Segmenting the video into discrete shots.
3. **Feature Extraction:** Generating multimodal embeddings and signals.
4. **Graph Construction:** Building the shot-node graph with temporal and semantic edges.
5. **GNN Importance Scoring:** Ranking shots based on graph-wide context.
6. **Selection & Merging:** Concatenating top-ranked shots while maintaining coverage and coherence.
7. **Text Summarization (Flan-T5):** Generating a readable, abstractive text summary from transcripts.
8. **Chapters & Evidence:** Grouping segments into navigable sections and providing explainable highlights.
9. **Delivery Layer:** Providing real-time feedback and exportable results via the UI.

6. System Architecture

VIDSUM-GNN follows a **pipeline + service** architectural style:

Major Components

- **Frontend (React + TypeScript):** An interactive dashboard for video configuration, real-time progress monitoring, and output visualization (merged video, text summary, evidence panel, and history).
- **Backend API (FastAPI):** Orchestrates the pipeline, streams progress via WebSockets, and manages the delivery of TTS (Text-to-Speech) and metadata.
- **AI/ML Inference Layer:** Performs the heavy lifting, including shot detection, feature extraction, GNN scoring, and Flan-T5 summarization.

- **Storage Layer:** Manages processing metadata and large binary artifacts like merged videos and audio files.

7. Results and Achievements

Functional Results

- **Condensed Video:** A coherent "Important Shots Compilation" is generated and playable.
- **Configurable Text Summary:** Supports multiple formats (bullet/structured/plain) and lengths (short/medium/long).
- **Navigational Chapters:** YouTube-style time-indexed sections for quick jumping.
- **Read-Aloud Support:** Integrated TTS with neural voice fallbacks for accessibility.

Interpretability Results

- **Evidence-Linked Highlights:** The system provides traceable justifications, connecting model selection to specific signals (e.g., high motion or transcript density) and graph relationships. This transforms the AI from a "black box" into a transparent tool.

Usability & Robustness

- **Progressive Feedback:** Real-time stage-based logs and progress bars ensure transparency during long processing tasks.
- **System Resilience:** Integrated TTS provider fallbacks (edge-tts and gTTS) and artifact caching ensure the system remains functional even under varied environmental conditions.
- **Clean Presentation:** Refined UI with scroll containment and normalized chapter titles for an efficient consumption experience.

8. Conclusion

VIDSUM-GNN represents a shift from extractive video processing to structural reasoning. By combining Graph Neural Networks with multimodal perception and abstractive language models, the system provides a high-fidelity summary that is not only shorter but also explainable and navigable.